

Arden Monaghan

780-361-8221 | scientificardenmonaghan@gmail.com | linkedin.com/in/arden-monaghan | [github](https://github.com)

EDUCATION

University of Alberta

Bachelor of Science in Computer Science, Specialization

Expected Graduation: Dec 2026

Edmonton, AB

- GPA: **3.7**
- **Relevant Coursework:** Data Structures and Algorithms, Introduction to Software Development, Machine Learning, Databases & SQL, Statistics, Computer Organization and Architecture, C Programming.
- **Relevant Awards:** Deans Research Award 2025, Jason Lang Scholarship, Deans Honor Roll x2

RELEVANT EXPERIENCE

Software Teaching Assistant

University of Alberta

May 2025 – Present

Edmonton, AB

- Guided over **200** students in learning **Software Engineering tools** by assisting labs teaching **Java development with JUnit** testing, **Github** workflows, and **NoSQL Firebase** integration.
- Mentored **4 Java development teams of 6 students** through **Software Development life-cycle** by implementing **Agile methodologies and Scrum standards**, increasing **software quality by over 50%**.

Software Developer

SAIL

May 2025 – Present

Calgary, AB

- Managed the development of Syncore, leading a team of **5 developers** using **Supabase and FastAPI**. Conducted tests in **3 clinics** and achieved a **48% reduction** in **patient query time** for **20+ clinicians**.
- Built **Python-based unit tests** and **Github Actions CI/CD pipeline** to publish **Docker images to AWS ECS** service, reducing team **integration errors** and **deployment time by over 65%**.

ML Research Intern

University of Alberta Electrical and Computer Engineering Faculty

Jan 2025 – Apr 2025

Edmonton, AB

- Implemented a **Camera Calibration system in OpenCV** to estimate **3D hand-joint positions** using ArUco Marker detection and camera-pose estimation, retrieving average offset error of **5%** from calculated positions.
- Integrated **SAM technology** using **PyTorch** to correctly compare 3D hand meshes to ground truth 2D masks.
- Fine-tuned model on manually created **dataset of 50 unique hand-object interactions** with applied forces.

Front-End Developer

LitLife Co.

Nov 2024 – Mar 2025

Edmonton, AB

- Facilitated the transition from Firebase to Supabase backend. Mapped **NoSQL to SQL schema** and reintegrated Flutterflow widgets, ensuring **accurate login functionality and profile displays**.
- **Revised user authorization** using **row-level security** systems in Supabase to reduce harmful database access from unregistered users, increasing application security by over **30%**.

PROJECTS

Moodify | *Android Studio, Java, Firebase, Git, Figma*

Jan 2025 – Apr 2025

- **Developed** home feed filtering, user authentication, and personal history **UI/UX** with **Firebase Cloud Storage** for a **Twitter-style Android app**, integrating queries for **displaying posts** from followed users in **real-time**.
- **Debugged and optimized** UI functionality and queries for mood creation, user-following, and user history features using **JUnit** and **Firebase Test Lab**, reducing data display errors by **25%**.

MentorMatch UAlberta | *Python, Flask, PyQt6, Docker, Ollama*

Apr 2024 – Sept 2024

- Automated the task of matching Mentors and Mentees using an **LLM-integrated algorithm** for the **Science Mentorship Program** at UAlberta, reducing time spent matching from **10 days to 10 minutes**.
- Delivered a secure PyQt GUI desktop app that integrated Llama3 embeddings with Mentor matching criteria. Enhanced overall program **operations by over 50%** while allowing **student data to remain private**.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C, C++, Dart, HTML, CSS, SQL, Swift, RISC-V

Libraries: SkLearn, NetworkX, Pandas, OpenCV, Pytorch, TensorFlow

Technologies/Frameworks: React, Supabase, Firebase, MongoDB, PostgreSQL, QT, Django, FastAPI, SQL Server

Developer Tools: Git, Docker, AWS, Azure, Cloud Computing